

GUGULOTHU AKHIL KUMAR

AI & Full-Stack Engineer

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github.com/LaZy-Wolf | Portfolio: akhil-kumar-portfolio.vercel.app

Summary

AI and full-stack engineer with 10+ months taking generative AI products from prototype to production deployment. Builds LLM agents, RAG pipelines, fine-tuned models and real-time voice AI with Python, FastAPI, LangGraph, PyTorch and Next.js, and measures each one against a benchmark.

Experience

GenAI Full-Stack Engineer Intern

Nov 2025 – Present

Allcognix AI

Remote, India

- Built and shipped 4 production AI products: **Vaelora** (AI sales engineer), **Graylint AI** (automated software QA), **Anvaya CRM** (AI-native CRM) and **Meeting AI** (self-hosted meeting transcription).
- Developed Vaelora's multi-agent orchestration: plain-English requests routed to 9 specialist agents and 60+ tools, integrated with HubSpot, Gmail, Google Calendar, Slack and Salesforce.
- Built Graylint AI's 6-stage agent pipeline that turns plain-English requirements into Playwright tests with self-healing locators and a CI gate, validated by mutation testing.
- Built Meeting AI on LiveKit, Deepgram and PostgreSQL with per-speaker audio channels; integrated LLMs into FastAPI and React/Next.js services with PostgreSQL, Redis and LangGraph, working in an agile team.

Projects

SONAR-TUNE: Fine-Tuned 2B Tool-Calling LLM | *PyTorch, Unsloth, LoRA, llama.cpp* | Case study 2026

- Fine-tuned Qwen3.5-2B with LoRA to replace SONAR's hosted 20B model (OpenAI gpt-oss-20b): **91.8% vs 67.2%** task success on a 61-case multi-turn benchmark with real tool execution ($p = 0.001$); 91.9% vs 45.9% exact tool arguments.
- Built an execution-validated synthetic data pipeline with a cross-family LLM judge (rejected about half of 1,240 conversations); 75.3% on a 150-case held-out benchmark, up from 45.3%.
- Fixed a chat-template train/serve mismatch; deployed on CPU via llama.cpp with 4-bit quantization (Q4_K_M GGUF, 1.3 GB) at \$0 per request.

SONAR: Real-Time Voice AI Agent | *LiveKit, WebRTC, MCP* | sonar-voice-agent.vercel.app Sep 2026 – Present

- Interruptible voice agent over browser (WebRTC) and phone (Twilio SIP) that books appointments through 6 tools on a custom MCP server; streaming Deepgram speech-to-text, Groq LLM, Cartesia text-to-speech.
- Published a per-stage latency budget: **1412 ms p50** time to first audio, 312 ms LLM first token; cut call pickup silence from 5.2 s to 1.64 s; 59 unit tests, GitHub Actions CI.

ASOC: Agentic RAG Helpdesk Copilot | *LangGraph, FastAPI, Qdrant, MCP, Langfuse*

2026

- Hybrid retrieval (dense semantic search + BM25, reciprocal rank fusion, cross-encoder rerank): **0.932 Recall@5** vs 0.864 dense-only on a 91-question golden set; human-in-the-loop approval before every write.

Technical Skills

Languages: Python, TypeScript, JavaScript, SQL

AI / ML: LLM agents, agentic AI, RAG, LangGraph, LangChain, MCP, tool / function calling, prompt engineering, fine-tuning (LoRA, PEFT), Hugging Face Transformers, PyTorch, LLM evaluation and observability (Langfuse), embeddings, vector databases (Qdrant, pgvector), multimodal AI, NLP, machine learning, deep learning

Voice AI: LiveKit Agents, WebRTC, Deepgram, Whisper, text-to-speech, Twilio SIP

Backend and Frontend: FastAPI, Flask, Django, Node.js, REST APIs, Next.js, React, Tailwind CSS

Data and Cloud: PostgreSQL, MySQL, MongoDB, Redis, SQLite, AWS (Lambda, S3), Docker, GitHub Actions CI/CD, Git

Education

B.Tech, Computer Science and Engineering (AI and Machine Learning)

2022 – 2026

Malla Reddy University CGPA: 8.6 / 10

Hyderabad, India

Achievements and Certifications

- 2nd Prize, National Hackathon, Malla Reddy University; Second-Round Qualifier, Google GRID Hackathon; Bolt AI Hackathon.
- Certifications: Google Cloud Computing (GDSC, 2024); Programming in Java (NPTEL, 2024).